

Pattern Recognition Course Project

Lightweight Multimodal Document Retrieval with ColSmolVLM

Model Scaling, Late Interaction, and Token Compression for Visual RAG

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Roll number: _____

Project scope. This report studies the retrieval stage of a multimodal retrieval-augmented generation (RAG) system. It evaluates released ColSmolVLM checkpoints; it does not claim to train the models or evaluate an answer generator.

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Abstract

A retrieval-augmented generation system can answer from a document only after its retriever finds the right evidence. That first step becomes fragile when a PDF communicates through layout, tables, plots, logos, and spatial position rather than plain text alone. This project therefore asks a focused pattern-recognition question: *how should a lightweight system represent and compare visually rich document pages?* We reproduce a page-level retrieval pipeline on four ViDoRe tasks and compare OCR-based BM25, dense BGE text retrieval, a single global visual vector, and ColSmolVLM multi-vector late interaction. We also compare 256M and 500M checkpoints and test four token-compression strategies at five retention levels. Across DocVQA, InfoVQA, ArxivQA, and TatDQA, 500M MaxSim obtains nDCG@5 of 57.8, 86.9, 74.7, and 76.5, respectively, outperforming every tested baseline on every dataset. Replacing MaxSim with global pooling loses 38.6–54.1 nDCG points, showing that the representation and matching rule matter more than the modest increase from 256M to 500M. K-means retains at least 95% of uncompressed quality with only 12.5–25% of page tokens, reducing index memory by 4–8×. The result is a compact, reproducible retrieval study—the scientific core needed before adding a generator to a visual RAG application.

1 The story begins before generation

Suppose a user asks, “What does the colour gradient in this scientific figure represent?” A language model may be fluent, but the answer is not necessarily stored in its parameters, and those parameters do not provide a trustworthy link to the user’s page. RAG addresses this by treating an external collection as memory [1]. In its simplest form,

$$\hat{a} = G(q, \text{TopK}_{D \in \mathcal{C}} S(q, D)), \quad (1)$$

where q is the question, $\mathcal{C}$ is the document collection, S is a retrieval score, G is a generator, and $\hat{a}$ is the answer. Equation 1 exposes the dependency that motivates this project: if the relevant page is absent from the top- K , the generator never sees the evidence. Better wording cannot repair missing evidence.

The usual document pipeline is

$$\text{PDF} \rightarrow \text{OCR} \rightarrow \text{text chunks} \rightarrow \text{embeddings} \rightarrow \text{nearest neighbours}.$$

It is practical, but OCR turns a two-dimensional visual object into a one-dimensional string. A table’s row–column relationship, a plot legend, small corner text, and typographic hierarchy may be lost or scrambled. Errors then propagate: OCR failure changes the text; chunking changes the context; a text encoder cannot recover visual information that no longer exists.

This project studies the alternative:

$$\text{page image} \rightarrow \text{visual token embeddings} \rightarrow \text{late interaction} \rightarrow \text{relevant pages}.$$

The generator is intentionally outside the experimental boundary. The goal is not to build another chatbot interface, but to understand the representation that would supply a future generator with evidence.

1.1 Research questions

The experiments answer four concrete questions.

1. Does direct visual retrieval beat OCR-based lexical and dense-text baselines?
2. Is a set of local page vectors better than one globally pooled vector?
3. Does the larger 500M ColSmolVLM checkpoint justify its extra size over 256M?
4. How many page tokens can be removed before retrieval quality collapses?

2 Background and related work

RAG. Lewis et al. combine a parametric sequence model with non-parametric retrieved memory, making knowledge easier to update and evidence easier to trace than in a purely parametric model [1]. Our work isolates the retrieval term $S(q, D)$ in Equation 1; answer quality is not measured.

Text retrieval. BM25 is a strong sparse baseline that rewards exact term overlap while normalising for document length [2]. Dense bi-encoders instead map a query and a document to semantic vectors and compare them by cosine similarity. We use BGE-small-en-v1.5 as a compact dense baseline [3].

Late interaction. ColBERT encodes query and document tokens independently, then performs a fine-grained MaxSim comparison at retrieval time [4]. This preserves local evidence while allowing document vectors to be computed offline.

Visual document retrieval. ColPali transfers late interaction to rendered document pages: a vision-language model emits multiple page vectors directly from the image, avoiding an OCR-first representation bottleneck [5]. The same work introduces ViDoRe, a benchmark spanning visually rich page-retrieval tasks. ColSmolVLM applies the idea to compact SmolVLM backbones; the released checkpoints are designed for multi-vector retrieval [6, 7].

3 Theory: from a page to a relevance score

3.1 The retrieval problem

For query q_i and corpus $\mathcal{C} = \{D_1, \dots, D_N\}$, the retriever returns a permutation π_i such that

$$S(q_i, D_{\pi_i(1)}) \geq S(q_i, D_{\pi_i(2)}) \geq \dots \geq S(q_i, D_{\pi_i(N)}). \quad (2)$$

The pattern-recognition problem is therefore a combination of (i) choosing a representation $\phi(\cdot)$ and (ii) choosing a comparison rule $S(\cdot, \cdot)$. The experiments hold the evaluation collection fixed while changing these two choices.

3.2 Text baselines

For BM25, a query term t contributes

$$\text{BM25}(q, D) = \sum_{t \in q} \text{IDF}(t) \frac{f(t, D)(k_1 + 1)}{f(t, D) + k_1 \left(1 - b + b \frac{|D|}{\overline{|D|}}\right)}, \quad (3)$$

where $f(t, D)$ is term frequency, $|D|$ is document length, $\overline{|D|}$ is average length, and this implementation uses $k_1 = 1.5$, $b = 0.75$. BM25 is interpretable and excellent for exact names or numbers, but it relies on OCR and weakly models paraphrase.

The BGE bi-encoder produces one normalised vector for each OCR document and query:

$$S_{\text{text}}(q, D) = \frac{\mathbf{e}_q^\top \mathbf{e}_D}{\|\mathbf{e}_q\|_2 \|\mathbf{e}_D\|_2}. \quad (4)$$

It can match semantic paraphrases but still only sees the extracted text.

3.3 Global visual pooling

Let the visual encoder produce n page-token vectors $D = [\mathbf{d}_1, \dots, \mathbf{d}_n] \in \mathbb{R}^{n \times d}$. A simple baseline averages the tokens,

$$\bar{\mathbf{d}} = \frac{1}{n} \sum_{j=1}^n \mathbf{d}_j, \quad S_{\text{global}}(q, D) = \bar{\mathbf{q}}^\top \bar{\mathbf{d}}. \quad (5)$$

This is memory-efficient, but averaging can cancel rare local signals. A small label in a chart competes with every background and body-text patch for space in the same vector.

3.4 Multi-vector late interaction

ColSmolVLM returns m query vectors $Q = [\mathbf{q}_1, \dots, \mathbf{q}_m]$ and n document vectors $D = [\mathbf{d}_1, \dots, \mathbf{d}_n]$, each in $\mathbb{R}^d$. After L_2 normalisation, their dot product is cosine similarity. MaxSim scores the best matching page token for each query token and then sums:

$$S_{\text{MaxSim}}(Q, D) = \sum_{i=1}^m \max_{1 \leq j \leq n} \mathbf{q}_i^\top \mathbf{d}_j. \quad (6)$$

The maximum gives each query concept its own location on the page. The sum requires a document to explain the query as a whole.

Worked example. Imagine three query tokens—“net”, “income”, and “2019”—and four page regions. Their similarity matrix is

$$QD^\top = \begin{bmatrix} 0.92 & 0.10 & 0.08 & 0.21 \\ 0.12 & 0.88 & 0.14 & 0.18 \\ 0.07 & 0.19 & 0.91 & 0.15 \end{bmatrix}. \quad (7)$$

Taking the row maxima yields $[0.92, 0.88, 0.91]$, so $S = 2.71$. The three concepts may occur in different cells of a table; MaxSim preserves that fact. A single average vector can blur the same three local matches.

The exact scoring cost per query–page pair is $O(mnd)$, and a half-precision page index requires approximately

$$M_{\text{index}} = N n d \times 2 \text{ bytes}. \quad (8)$$

This explains the central trade-off: multi-vector matching improves recognition, but token count controls storage and arithmetic.

3.5 Token compression

For retention ratio $r \in \{1, .75, .50, .25, .125\}$, the target token count is $k = \max(1, \lceil rn \rceil)$. We compare:

- **random:** sample k tokens without replacement (three seeds);
- **uniform:** retain approximately evenly spaced token positions;
- **mean-pool:** divide the sequence into k contiguous groups and average each group;
- **k-means:** replace the tokens with k learned centroids.

The first two select tokens, while the latter two construct representatives. K-means approximately minimises

$$\min_{\{\mathbf{c}_\ell\}_{\ell=1}^k} \sum_{j=1}^n \min_{\ell} \|\mathbf{d}_j - \mathbf{c}_\ell\|_2^2, \quad (9)$$

so it preserves the geometry of repeated visual patterns rather than merely preserving positions.

3.6 Evaluation measures

The primary metric is nDCG@5. For ranked gains g_i ,

$$\text{DCG}@K = \sum_{i=1}^K \frac{2^{g_i} - 1}{\log_2(i + 1)}, \quad \text{nDCG}@K = \frac{\text{DCG}@K}{\text{IDCG}@K}. \quad (10)$$

It rewards putting a relevant page near the top and normalises across queries. We also report mean reciprocal rank, $\text{MRR} = |Q|^{-1} \sum_q 1/r_q$, where r_q is the first relevant rank, and $\text{Recall}@K$, the fraction of queries with relevant evidence in the first K results.

4 Methodology and implementation

4.1 Datasets

We use the complete public test splits exposed through MTEB/ViDoRe, with no locally supplied PDFs. Table 1 shows the exact evaluation sizes. Each item is a natural-language query with page-level relevance labels.

Table 1: Evaluation collections and usable OCR coverage. Mapping coverage was 100% for all four datasets; “usable OCR” counts pages with non-empty extracted text.

Dataset	Queries	Pages	Usable OCR	Dominant visual content
DocVQA	451	500	496 (99.2%)	scanned forms and documents
InfoVQA	494	500	497 (99.4%)	infographics
ArxivQA	500	500	422 (84.4%)	scientific plots and figures
TatDQA	1,646	277	277 (100%)	financial tables and reports

The OCR pages are joined to the MTEB corpus by exact image digest (or filename when available). The code distinguishes a mapped-but-empty OCR row from a missing mapping. This matters most for ArxivQA: its text scores are valid for the supplied OCR artifact, but 78 pages contain no OCR text, so visual–text comparisons there should be interpreted cautiously.

4.2 Compared systems

Table 2: Experimental factors. All released neural checkpoints are evaluated without project-specific fine-tuning.

Family	Representation	Score	Purpose
BM25	Tesseract OCR term counts	BM25	lexical text baseline
BGE-small	one 384-D OCR-text vector	cosine	semantic text baseline
Global visual	one pooled 128-D page vector	dot product	visual single-vector ablation
ColSmol MaxSim	many 128-D page vectors	Eq. 6	proposed retrieval mechanism

The visual experiments compare `vidore/colSmol-256M` and `vidore/colSmol-500M`. The 500M checkpoint contains 477,663,552 parameters in the loaded implementation. All neural inference is frozen; no labels from the evaluation sets are used for training.

4.3 Reproducible execution

Experiments ran with seed 42 in Python 3.12.12, PyTorch 2.11.0+cu130, Transformers 5.17.0, and ColPali Engine 0.3.18 on one NVIDIA RTX A6000 (physical GPU 2, 47.4 GiB). FP16 is used for visual embeddings. Dataset fingerprints, checkpoint revisions, timings, per-query ranks, and environment information are stored with the results. Twenty automated unit tests passed in the final server run.

The implementation follows the mathematical structure directly:

Listing 1: Core late-interaction scorer (simplified from the project implementation).

```
# Q: [batch, query_tokens, dim]
# D: [page_tokens, dim]
similarities = Q @ D.T
score = similarities.max(dim=-1).values.sum(dim=-1)
```

The full experiment driver evaluates both checkpoints, all four datasets, two text baselines, 100 compression configurations, and then regenerates summary tables, plots, and failure cases. Existing valid caches are reused to make interrupted runs resumable.

5 Quantitative results

5.1 Main comparison

Table 3: nDCG@5 (%). Best result per dataset is bold. “Global” and “MaxSim” use page images; BM25 and BGE use OCR text.

Representation	Model	DocVQA	InfoVQA	ArxivQA	TatDQA
Text lexical	BM25	34.8	60.9	18.5	57.4
Text dense	BGE-small-en-v1.5	27.1	68.5	32.3	31.4
Global visual	ColSmol-256M	16.6	45.4	18.3	21.3
Global visual	ColSmol-500M	19.1	55.5	25.3	28.4
MaxSim visual	ColSmol-256M	55.6	83.6	72.4	75.7
MaxSim visual	ColSmol-500M	57.8	86.9	74.7	76.5

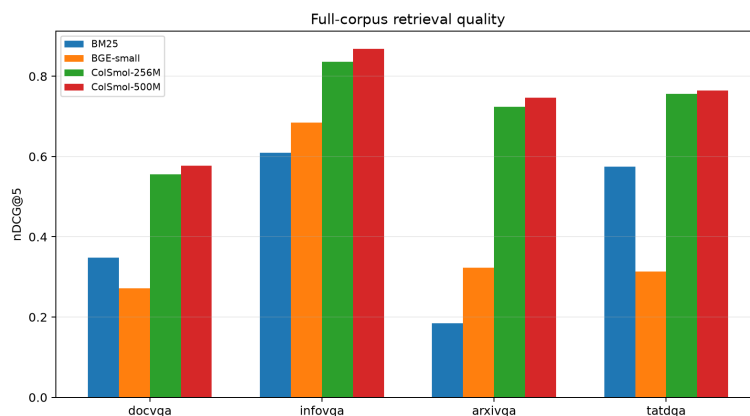


Figure 1: Main nDCG@5 comparison generated from the committed result JSON files.

Three observations tell the main story.

1. **Late interaction is the decisive component.** With the same 500M encoder, MaxSim exceeds global pooling by 38.6 points on DocVQA, 31.4 on InfoVQA, 49.4 on ArxivQA, and 48.1 on TatDQA. Local matching is therefore not a cosmetic scoring change; it prevents crucial page regions from being averaged away.
2. **Visual retrieval is consistently strongest.** MaxSim-500M beats the best text baseline by 23.0, 18.4, 42.4, and 19.0 points on the four datasets. The largest gap occurs on scientific figures, exactly where OCR has both a representational disadvantage and incomplete text coverage.
3. **Scaling helps, but less than representation.** Moving from 256M to 500M improves MaxSim by 2.2, 3.2, 2.2, and 0.8 points. By contrast, changing global pooling to MaxSim gives tens of points. For this scale, keeping local tokens is more important than doubling the checkpoint class.

For the 500M MaxSim system, Recall@1 / Recall@5 / Recall@10 is 50.0/64.3/71.5 on DocVQA, 82.4/90.5/92.7 on InfoVQA, 67.0/81.2/85.8 on ArxivQA, and 64.1/87.4/93.7 on TatDQA. Thus, most remaining errors are not only ordering errors: on DocVQA, 28.5% of queries still have no relevant page in the top ten.

5.2 Compression ablation

Table 4: Smallest configuration retaining at least 95% of the uncompressed nDCG@5. All selected configurations use k-means.

Dataset	Tokens kept	Base nDCG	Compressed nDCG	Base MB	Compressed MB
DocVQA	25%	57.8	56.6	129.36	32.36
InfoVQA	12.5%	86.9	83.4	87.08	10.84
ArxivQA	12.5%	74.7	71.8	107.02	13.35
TatDQA	25%	76.5	73.0	72.38	18.11

Equation 8 predicts the almost linear storage reduction visible in Table 4: retaining one eighth of the tokens gives roughly one eighth of the index. K-means is the strongest aggressive compressor because its centroids preserve occupied regions of embedding space. Uniform and random selection may discard rare but decisive tokens; contiguous mean pooling may mix unrelated neighbouring regions.

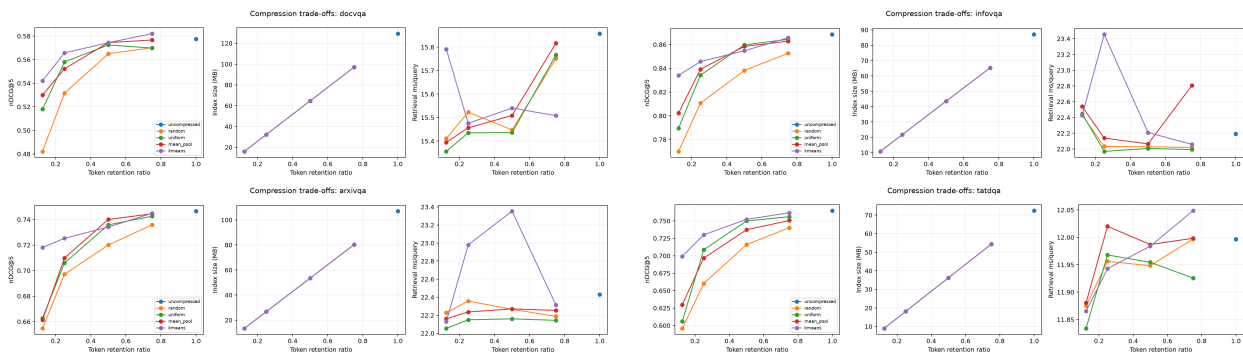


Figure 2: Quality-storage trade-offs for four compression methods. Random results aggregate three seeds; deterministic methods use one run.

Compression did *not* consistently reduce measured query latency. For example, DocVQA falls from 16.06 to 15.48 ms/query at its chosen point, whereas InfoVQA rises from 16.68 to 22.43 ms/query. At these corpus sizes, GPU launch, batching, padding, top- k , and sorting overhead can dominate

the number of dot products. The defensible conclusion is therefore strong memory savings with preserved quality, not a universal latency speed-up.

5.3 Computational requirements

Table 5: Measured 500M visual-system efficiency. Encoding peak VRAM is about 2.9 GiB on every dataset.

Dataset	Page encode (ms)	Query encode (ms)	MaxSim (ms/query)	Index (MB)
DocVQA	693.1	16.5	16.1	129.36
InfoVQA	500.6	16.4	16.7	87.08
ArxivQA	577.7	21.2	22.7	107.02
TatDQA	706.2	16.5	9.1	72.38

Page encoding is the expensive offline operation; query encoding and scoring are online. The index remains small here because the benchmark corpora contain only 277–500 pages. For a million pages with 1,000 tokens/page, $d = 128$, and FP16 storage, Equation 8 gives approximately 238 GiB, which motivates compression or specialised multi-vector indexing for deployment.

6 Qualitative analysis: where the numbers come from

Table 6 traces individual questions through the rankings. These are not generated explanations of page content; they are direct case studies from the stored per-query ranks.

Table 6: Representative retrieval cases. Lower rank is better.

Dataset	Query	MaxSim	Global	BM25	BGE
ArxivQA	What can be inferred about the term “alexa” in the given figure?	1	234	400	478
InfoVQA	What does “R” stand for?	1	239	293	430
TatDQA	What was the percentage change in the net total between 2018 and 2019?	1	243	137	177
InfoVQA	Which is element 67 on the periodic table?	44	23	1	1
DocVQA	What are the letters in the monogram logo at the top right?	447	497	45	339
ArxivQA	What does the figure primarily illustrate?	385	286	480	4

The first three cases show the advantage of local visual evidence. A term inside a plot, a one-letter infographic label, or a relationship spanning table cells can be matched without requiring OCR to linearise the whole page. The periodic-table case shows the opposite: a distinctive fact expressed cleanly in text is ideal for lexical or dense text retrieval.

The failures are equally informative. A tiny monogram in a named corner requires both fine visual resolution and explicit spatial grounding; neither representation reliably solves it. The generic question “What does the figure primarily illustrate?” contains few discriminative words, so many scientific pages can look semantically plausible. BGE ranks the relevant page fourth, demonstrating that visual and text channels make complementary errors rather than forming a strict hierarchy.

7 Discussion, limitations, and threats to validity

What was learned. The strongest conclusion is architectural: preserving a set of local descriptors and delaying interaction until the query arrives is dramatically more effective than compressing a page to one vector. The 500M model is the best tested system, but the 256M MaxSim model already beats every text and global baseline. K-means then recovers much of the storage cost without erasing the advantage.

OCR comparability. Text baselines depend on supplied Tesseract artifacts. Non-empty OCR coverage is 84.4% on ArxivQA, while the visual methods see all 500 page images. This is a real operational advantage of the visual route, but it also means that the ArxivQA gap cannot be attributed purely to representation quality. The other three datasets have 99.2–100% non-empty OCR coverage.

Scale. Full benchmark splits were used, but each retrieval corpus is small. Brute-force GPU MaxSim is reasonable for 277–500 pages and does not prove million-page scalability. Approximate multi-vector search, sharding, candidate generation, or reranking would be necessary at production scale.

No task-specific training. This is an evaluation and ablation project. It uses released ColSmol checkpoints and does not fine-tune on the four test sets. That choice gives a clean study of off-the-shelf representations, but does not answer how contrastive fine-tuning or parameter-efficient adaptation would change the trade-off.

No answer-generation evaluation. Although RAG motivates the work, the experiments stop after retrieval. There is no language-model answer, faithfulness metric, or hallucination study. Calling the current system a complete RAG pipeline would therefore be incorrect. The honest claim is that it implements and evaluates the retriever required by a future visual RAG system.

Qualitative evidence. Per-query successes and failures are analysed, satisfying the need for qualitative interpretation, but no token-to-region heatmaps were generated. Future work should overlay each query token’s argmax location from Equation 6 on the page. This would turn the ranking score into a spatial explanation and help distinguish encoder failures from ambiguous annotations.

8 From this retriever to a complete RAG system

The next engineering step is short, but its evaluation is a separate experiment:

1. index each page once with compressed ColSmol vectors;
2. retrieve the top- K page images with MaxSim;
3. give those images and the question to a vision-language generator;
4. require citations to page identifiers;
5. score retrieval and generation separately, including answer correctness and faithfulness.

This separation is important. Retrieval Recall@ K asks whether the evidence was available; answer accuracy asks whether the generator used it; faithfulness asks whether the answer is actually supported. Conflating the three makes a failure impossible to diagnose.

9 Conclusion

This project began with a simple fact: generation cannot use evidence that retrieval fails to find. For visually rich documents, OCR-first text retrieval discards part of the signal before matching begins. ColSmolVLM instead represents a page as many local visual vectors, and MaxSim lets each query concept find its strongest region. On four complete ViDoRe tasks, that design beats BM25, dense OCR-text retrieval, and global visual pooling. The largest gain comes not from model size but from retaining local structure. K-means compression then keeps at least 95% of baseline nDCG@5 with only one eighth to one quarter of the visual tokens. The resulting system is a lightweight, reproducible, and mathematically interpretable retriever—a strong foundation for multimodal RAG, while remaining explicit about what has not yet been generated or evaluated.

Implementation provenance and academic integrity

The ColSmolVLM checkpoints, ColPali/ColBERT concepts, BGE checkpoint, MTEB/ViDoRe datasets, and Tesseract OCR artifacts are prior work and are cited below. The project contribution is the reproducible evaluation implementation, verified OCR-to-corpus join, global-pooling ablation, 256M/500M comparison, token-compression study, efficiency measurement, aggregation, plots, and per-query failure analysis. No benchmark result is presented as model training performed in this project.

References

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A Artifact map

The report can be audited from repository artifacts rather than reconstructed from prose:

- `artifacts/results/visual/`: eight full visual evaluations;

- `artifacts/results/text/`: four OCR text-baseline evaluations;
- `artifacts/results/compression/`: 100 compression configurations and manifests;
- `artifacts/summary/main_results.csv`: 24 primary method–dataset rows;
- `artifacts/summary/compression_aggregate.csv`: 68 aggregated compression rows;
- `artifacts/summary/failure_analysis.csv`: 30 difficult queries;
- `artifacts/results/environment_diagnostic.json`: software and GPU provenance.

B Minimal reproduction command

The repository README documents every experiment individually. The complete tested pipeline is:

```
source rag/bin/activate
GPU_ID=2 bash scripts/run_all_experiments.sh
```